

Task 5: Exploratory Data Analysis (EDA)

Titanic Dataset — Data Analyst Internship

1. Objective

The objective of this EDA is to explore the Titanic passenger dataset using statistical summaries and visualizations, identify patterns and relationships, and communicate useful findings.

2. Tools Used

Python, Pandas, NumPy, Matplotlib and Seaborn.

3. Dataset Overview

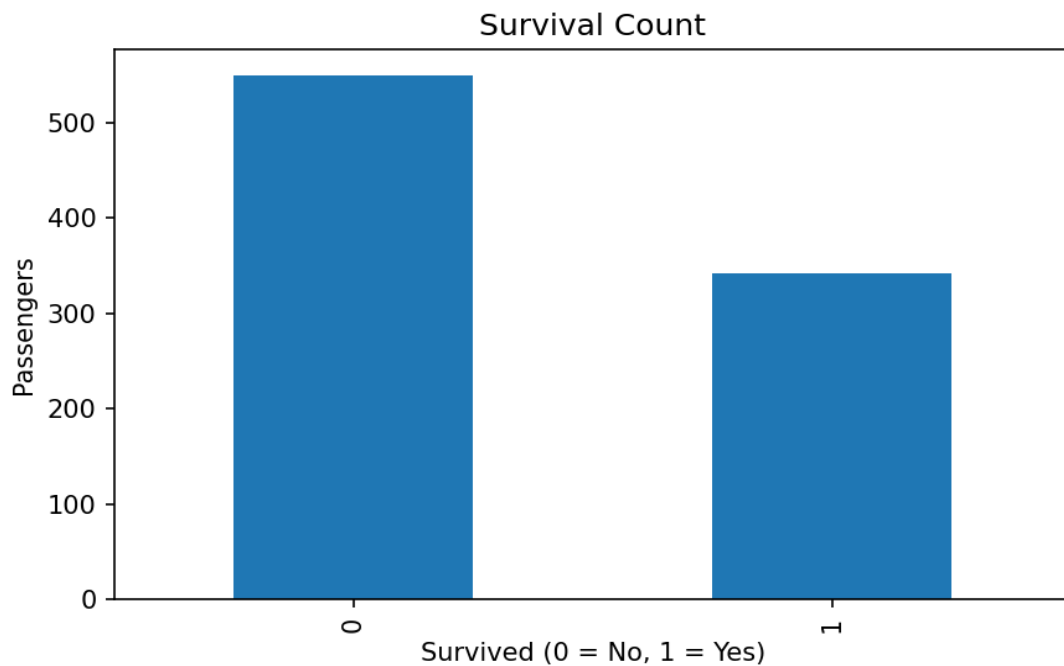
Item	Result
Rows	891
Columns	13
Target variable	Survived
Numerical variables	PassengerId, Survived, Pclass, Age, SibSp, Parch, Fare
Missing values	Age: 177

4. Statistical Findings

Measure	Value
Survived	342 (38.4%)
Did not survive	549 (61.6%)
Average age	29.70 years
Average fare	32.20
Female survival rate	74.2%
Male survival rate	18.9%
1st class survival rate	63.0%
2nd class survival rate	47.3%
3rd class survival rate	24.2%

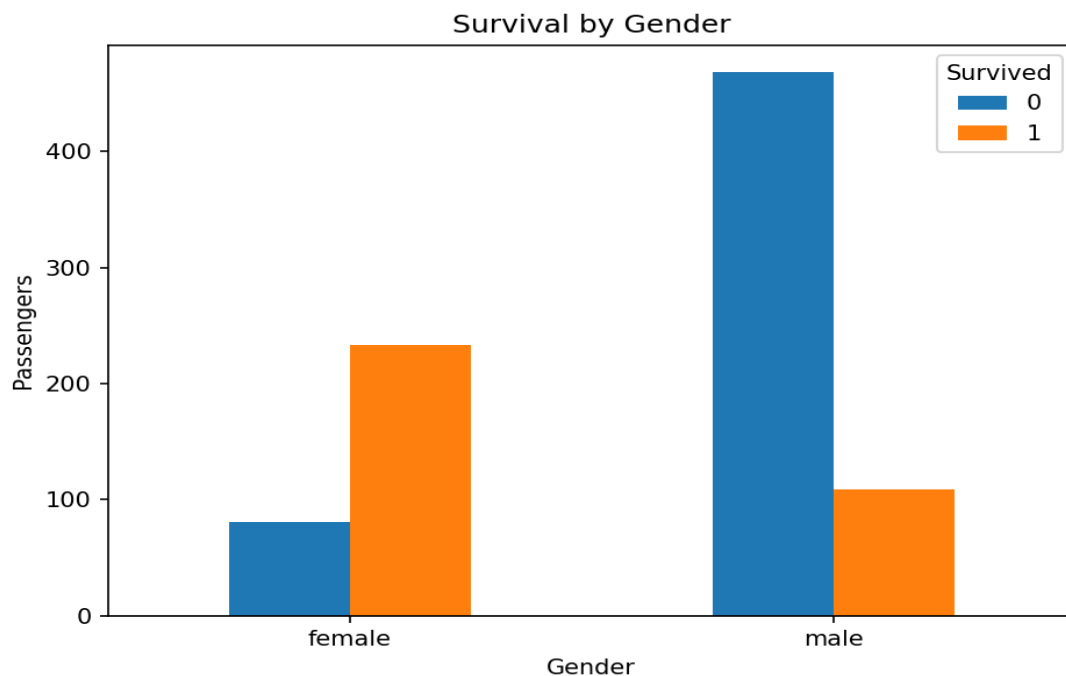
5. Visual Analysis and Observations

Survival Count



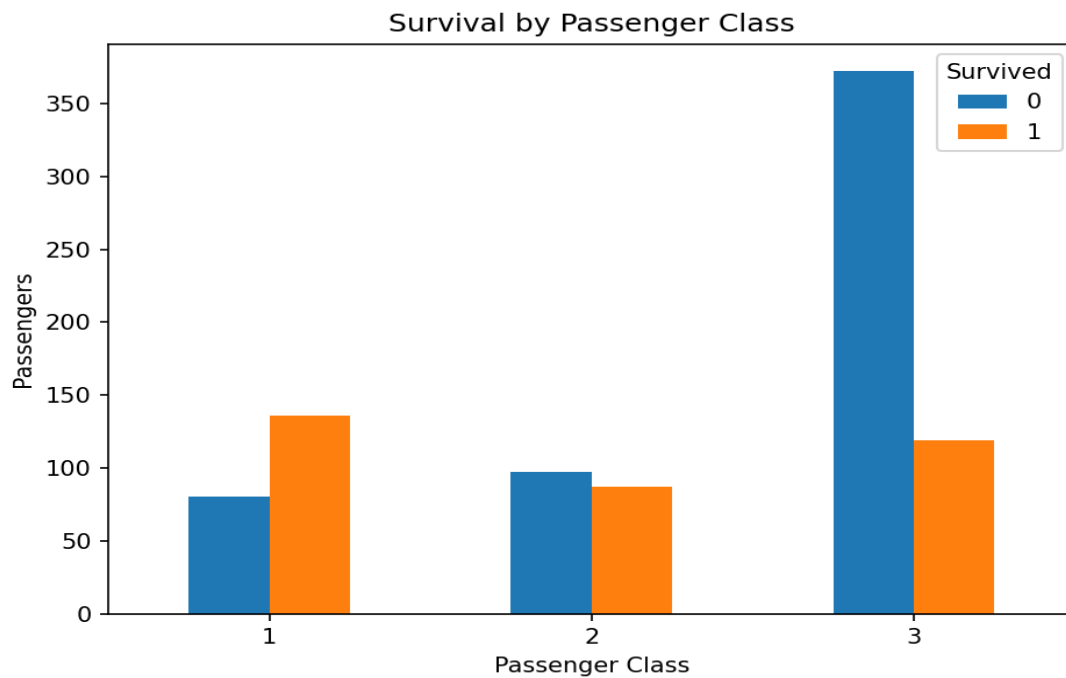
Observation: The dataset contains 342 survivors and 549 passengers who did not survive. The overall survival rate is approximately 38.4%.

Survival by Gender



Observation: Female passengers had a substantially higher survival rate (74.2%) than male passengers (18.9%). This indicates a strong relationship between gender and survival in this dataset.

Survival by Passenger Class



Observation: Survival varies by passenger class. The survival rates are approximately 63.0% for 1st class, 47.3% for 2nd class and 24.2% for 3rd class, showing better outcomes for higher classes.

6. Other EDA Visualizations

The Jupyter notebook also contains an age histogram, an Age vs Fare scatter plot, a correlation heatmap, an Age vs Survival boxplot, and a pairplot. These visualizations are used to examine distributions, relationships, correlations and possible outliers.

7. Conclusion

The EDA shows that survival on the Titanic was not evenly distributed. Gender and passenger class show clear relationships with survival, while age and fare provide additional information about passenger characteristics. Statistical summaries and visualizations together make these patterns easier to identify and communicate.

8. Internship Task Completion

The analysis uses the requested Python EDA workflow: Pandas for data handling, statistical functions such as `info()`, `describe()` and `value_counts()`, and Matplotlib/Seaborn visualizations including count plots, histogram, scatter plot, boxplot, heatmap and pairplot.